

Sandwich Bread Supply Chain Simulation Game

Toast & Mellow

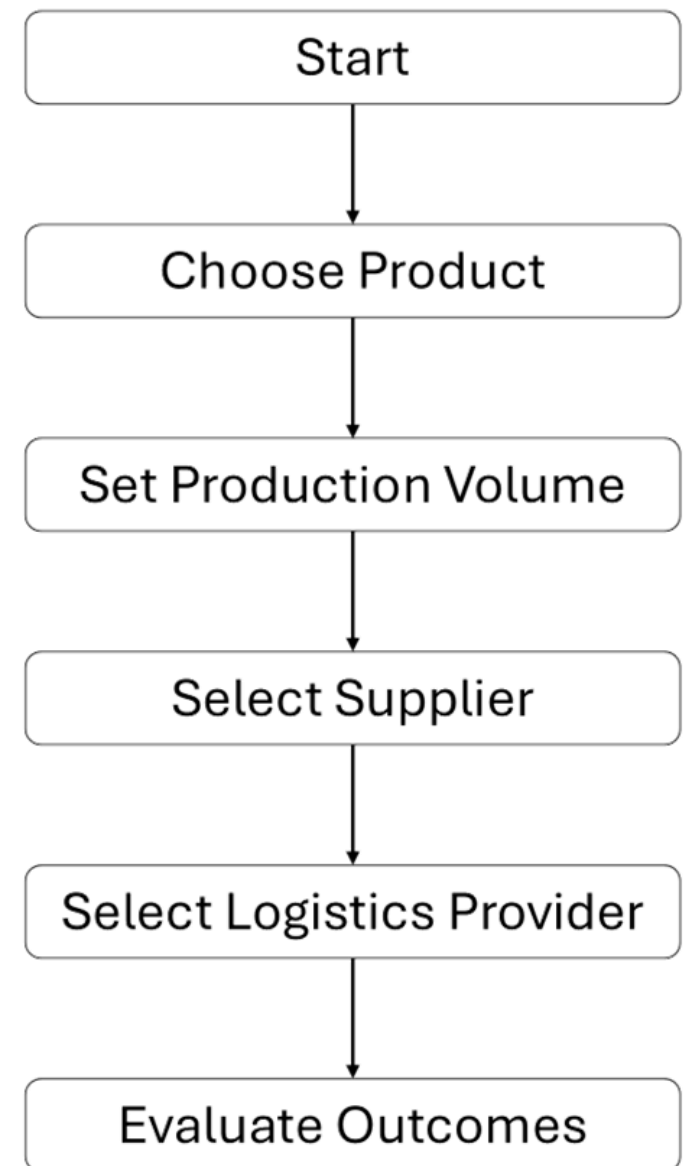


HARVEST GRAIN LOAF

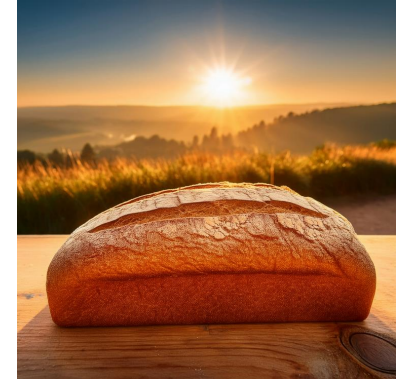


Overview

- Work in teams of four
- Decide:
 - Product choice
 - Production quantity
 - Supplier selection
 - Logistics service provider selection
- To improve sustainability and profitability



Product Choice



HARVEST GRAIN LOAF



Product Choice

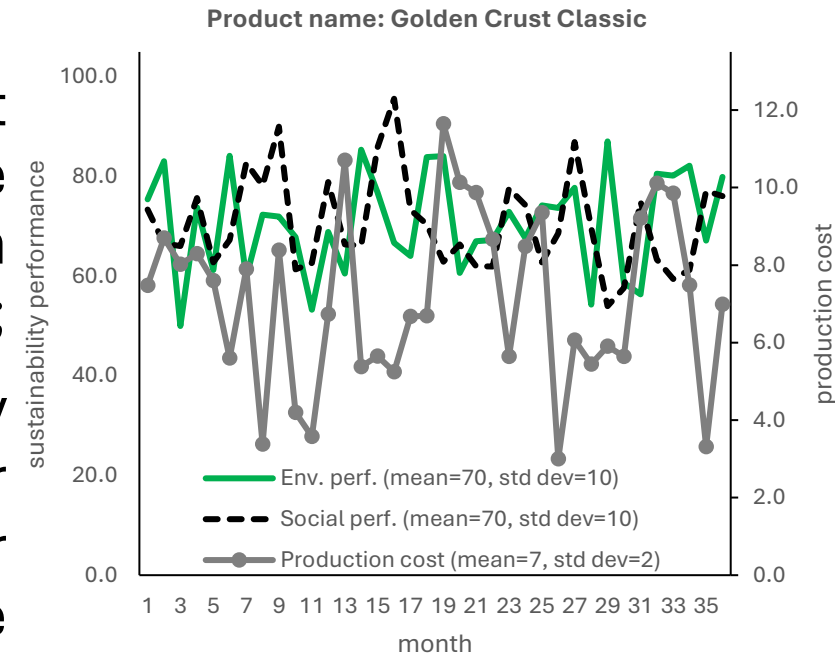
- Price:
 - Depends on customer willingness to pay
 - Influenced by the sustainability of the product and its delivery
 - Max: £110
- Your product choice affects revenue and branding
- Discuss with your management team carefully

Product 1: Golden Crust Classic



Soft and
fluffy;
Golden-
baked crust;
Comforting
bite

Low yet
unstable
production
cost;
Sustainability
is neither
high nor
stable



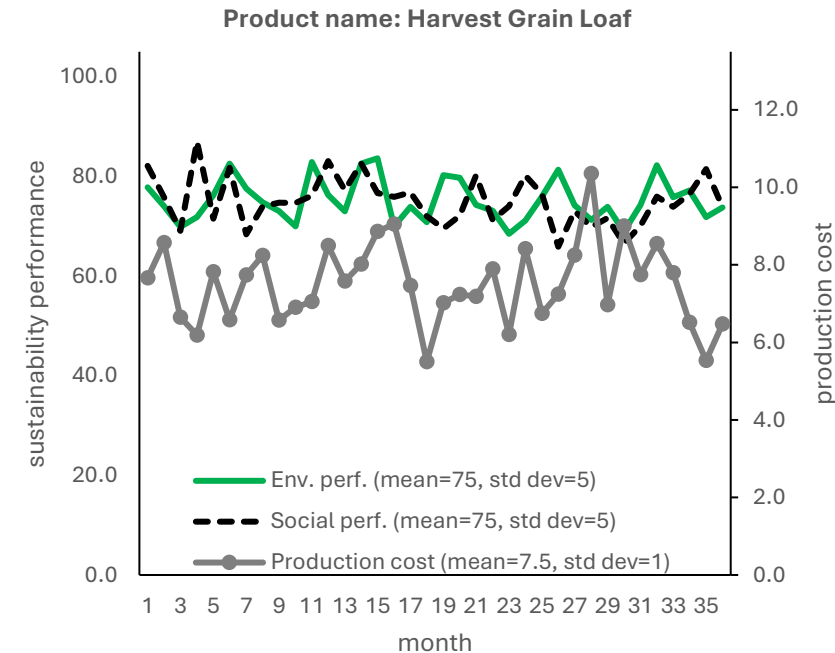
Product 2: Harvest Grain Loaf



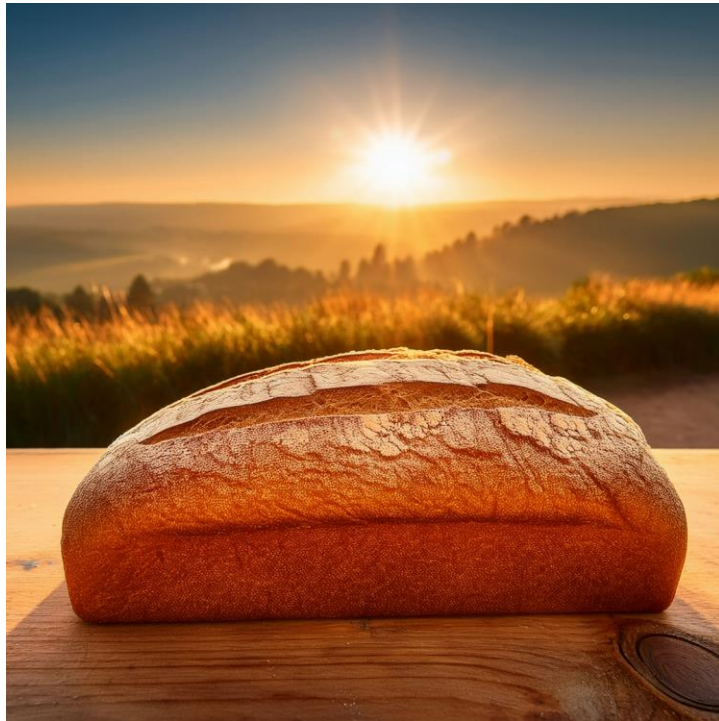
HARVEST GRAIN LOAF

A blend of
whole grains
and seeds;
Nutty
flavour;
Chewy
texture

Production
cost slightly
higher yet
more stable
than Golden
Crust
Classic;
So is its
sustainability

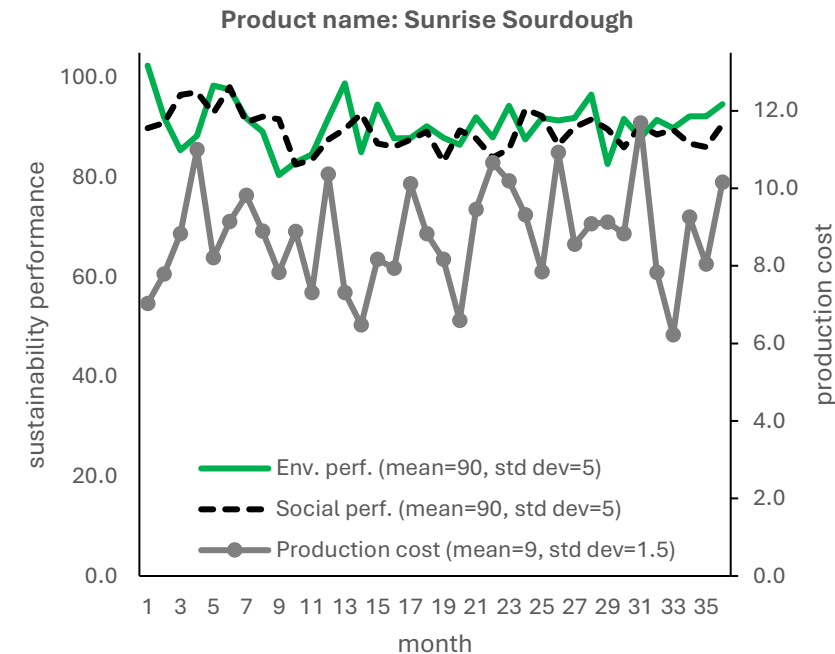


Product 3: Sunrise Sourdough



A naturally-leavened loaf;
Tangy depth of flavour;
Open crumb

Production cost higher yet less stable than Harvest Grain Loaf;
High yet unstable sustainability

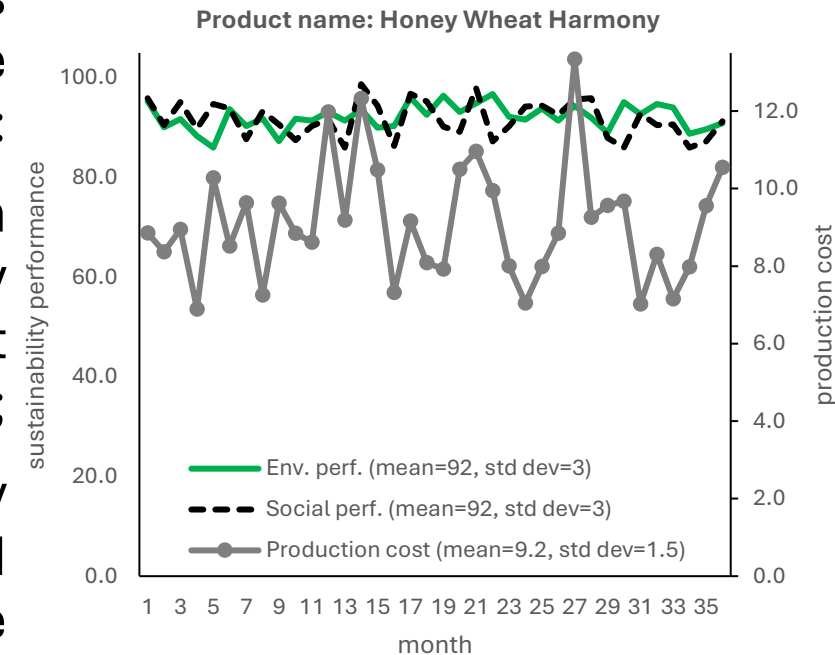


Product 4: Honey Wheat Harmony



Smooth and
subtly
sweet;
Whole
wheat flour;
Natural
honey;
Soft crumb

Vis-à-vis
Sunrise
Sourdough:
Production
cost slightly
higher yet
equally stable;
Sustainability
is higher and
more stable

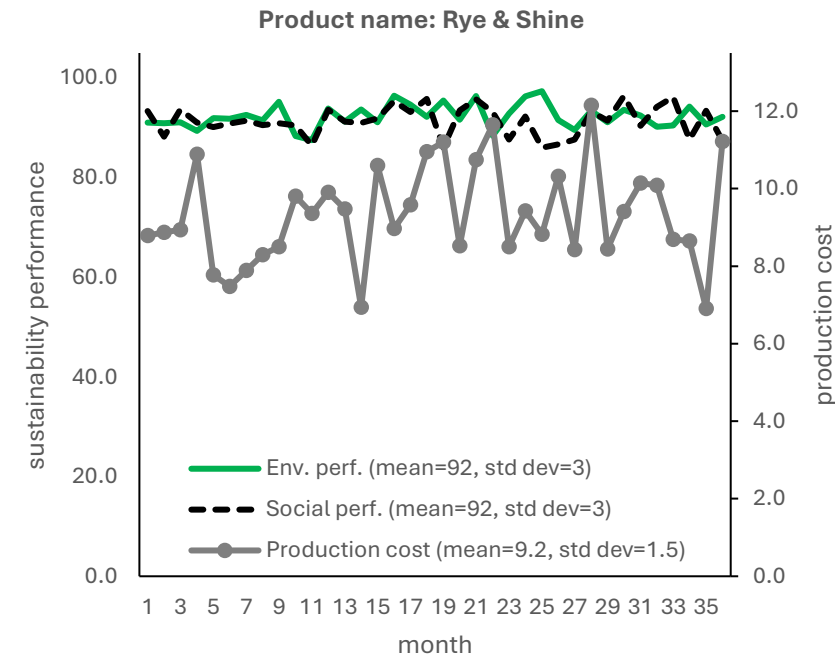


Product 5: Rye & Shine



Bold and
flavour-
forward;
Stone-
ground rye
flour;
Dense
texture and
robust taste

The
production
cost and
sustainability
outcomes are
comparable
to Honey
Wheat
Harmony

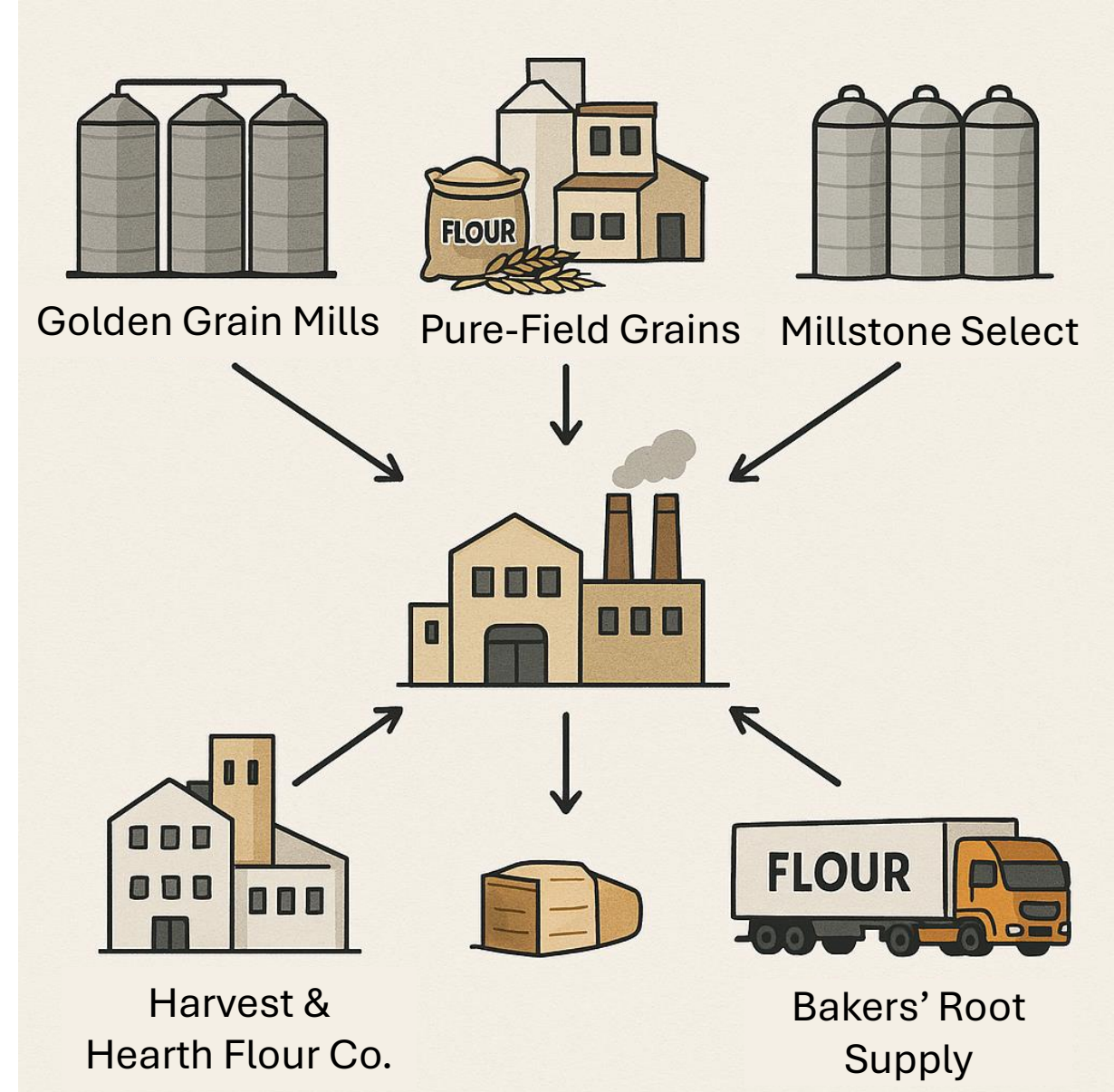


Production Quantity Determination

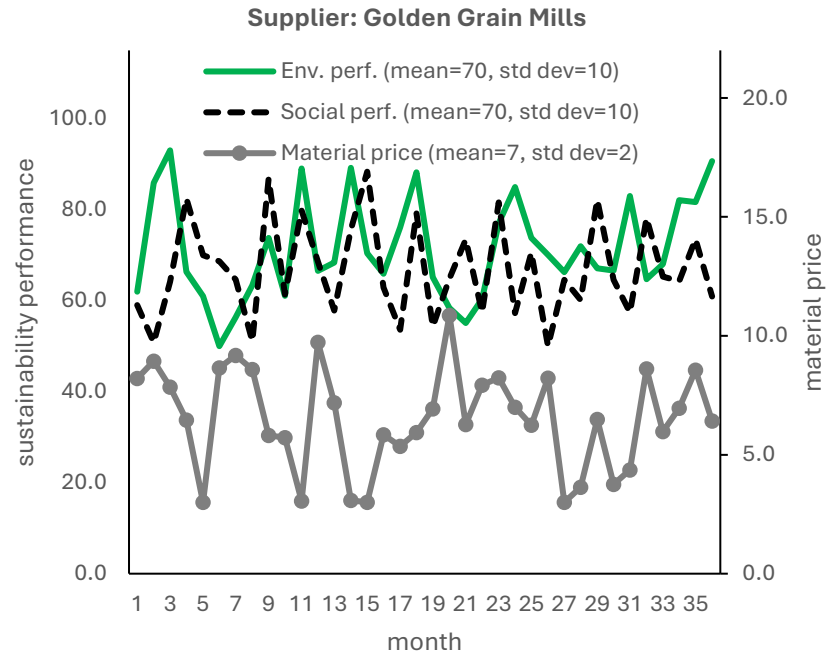
- Average monthly demand = 100, but the actual demand may differ significantly
- Production costs: production processes (including labour), materials procurement and logistics
- Production < Demand: low revenue
- Production > Demand: disposal cost of £10/unit + wasted production costs
- What is the optimal production volume? Discuss with your management team.

Supplier Selection

- Five material suppliers available
 - Price
 - Sustainability performance
 - Labour and machinery
- Which one should be selected to meet Toast & Mellow's organisational goals?
- Your team should review their historical performance.



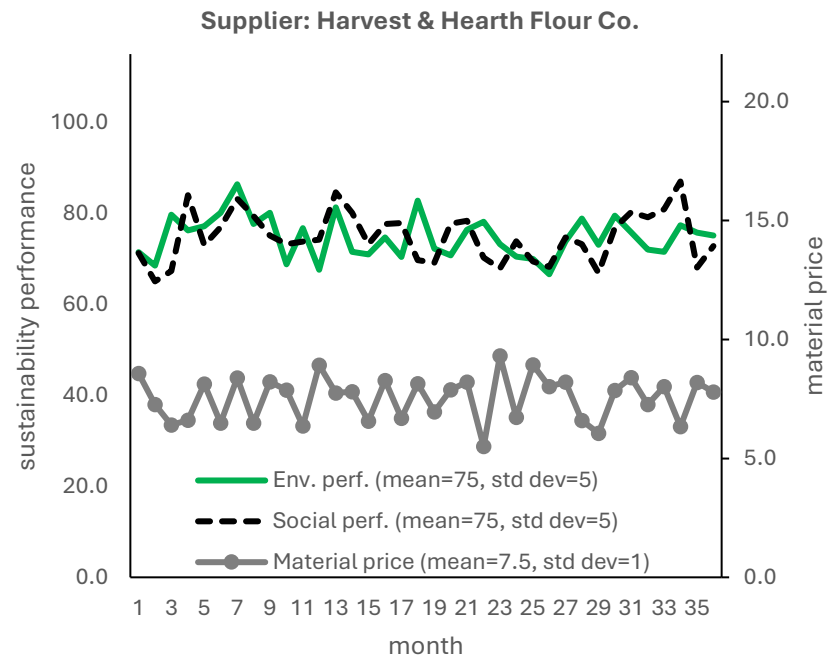
Supplier 1: Golden Grain Mills



- A large scale, industrial and highly-automated facility
- Sourcing conventional wheat from high-yield farms
- Competitive pricing
- Less advanced sustainability practices



Supplier 2: Harvest & Hearth Flour Co.

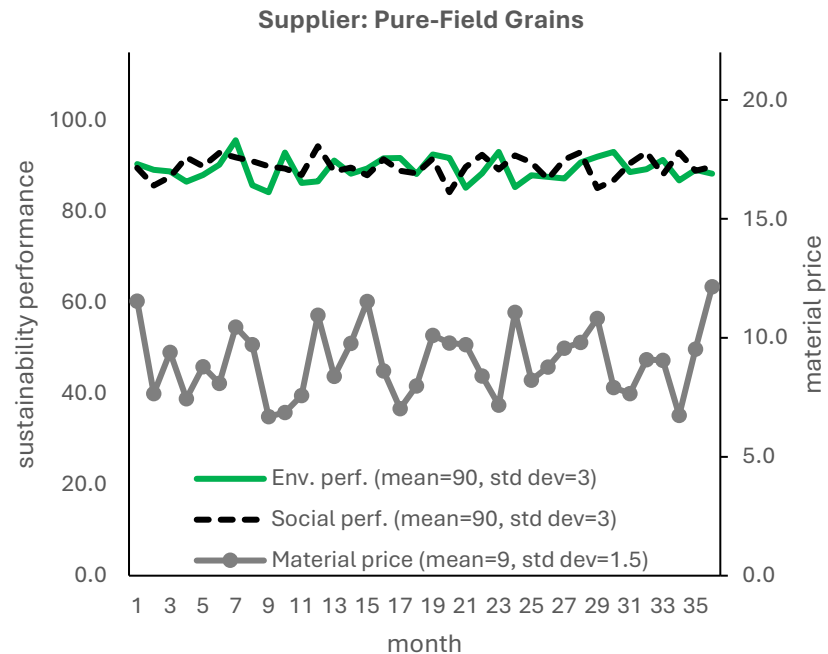


Compared to Golden Grain Mills:

- Slightly higher yet more stable prices
- Slightly higher and more stable sustainability outcomes
- Regional sourcing



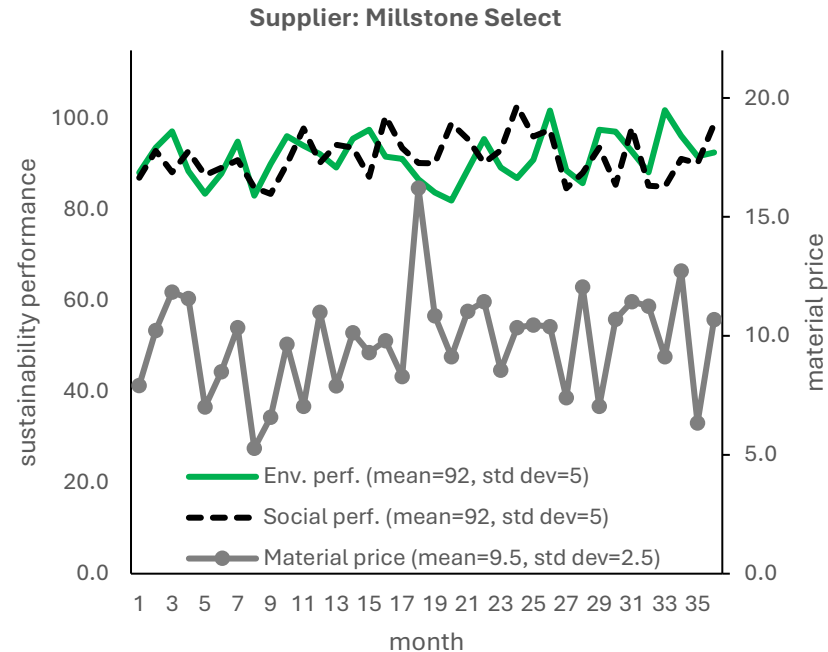
Supplier 3: Pure-Field Grains



- Exclusive sourcing from certified organic farms
- Regenerative farming
- High price
- High sustainability performance



Supplier 4: Millstone Select



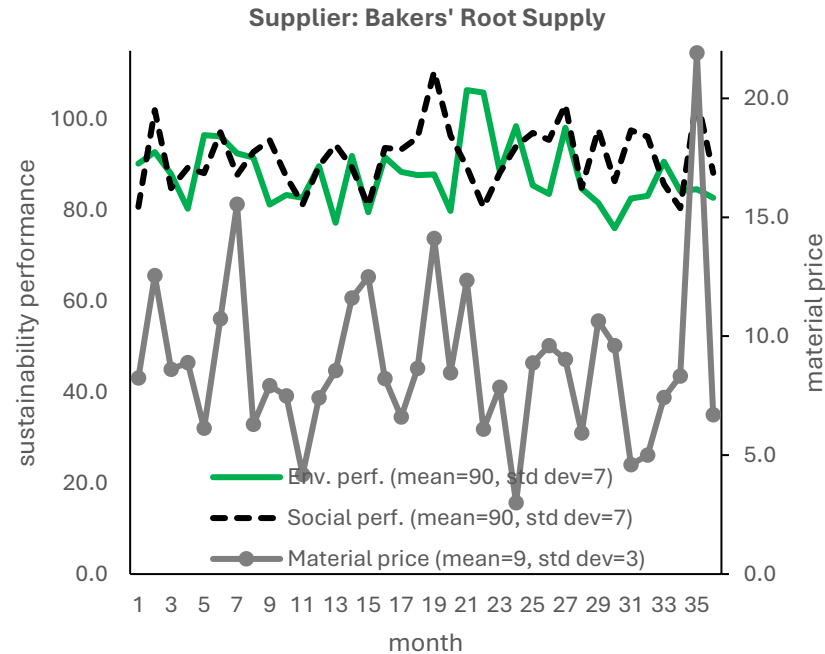
- Highest-quality, ethically sourced grains

Compared to Pure-Field Grains:

- Slightly higher yet more unstable prices
- Slightly higher yet less stable sustainability performance



Supplier 5: Bakers' Root Supply



- Community-based farming networks (decentralised)
- Compared to Pure-Field Grains:
- Equally high yet more unstable prices
 - Equally high yet less stable sustainability performance



Logistics Services

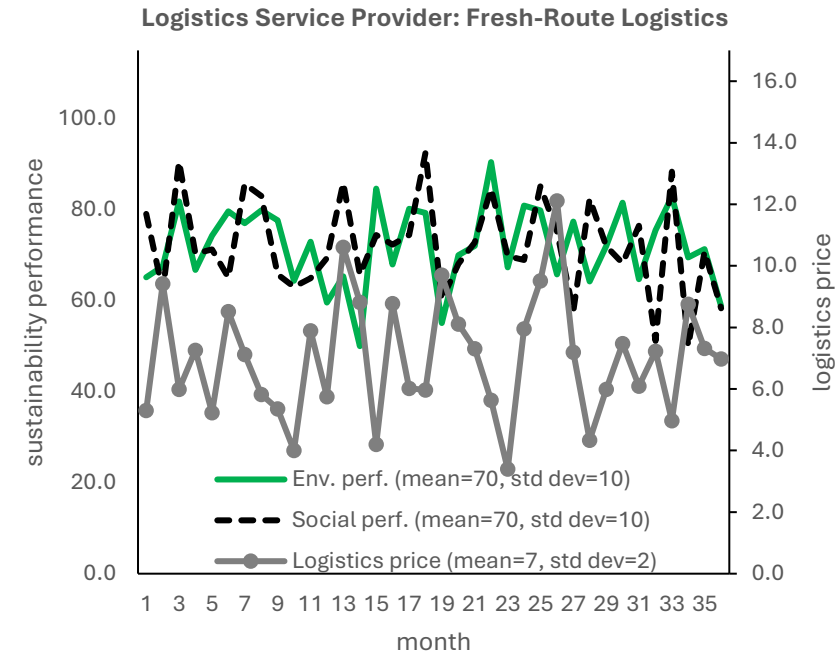
- Five logistics service providers available
 - Price
 - Sustainability performance
 - Labour and machinery
- Which one should be selected to meet Toast & Mellow's organisational goals?
- Your team should review their historical performance.



Logistics Service Provider 1: Fresh-Route Logistics



- Budget-friendly
- Fast turnaround and basic delivery
- Cost-efficiency
- A large fleet of conventional diesel vehicles



Logistics Service Provider 2: Crate & Freight Co.

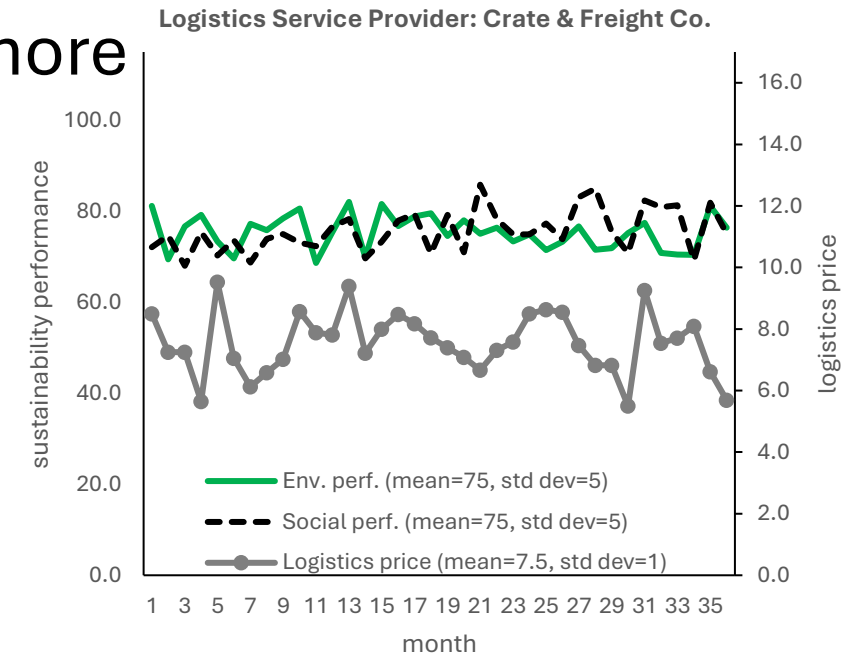
- Regional hub



- Partial fleet electrification

Compared to Fresh-Route Logistics

- Slightly higher yet more unstable prices
- Slightly higher and more stable sustainability performance

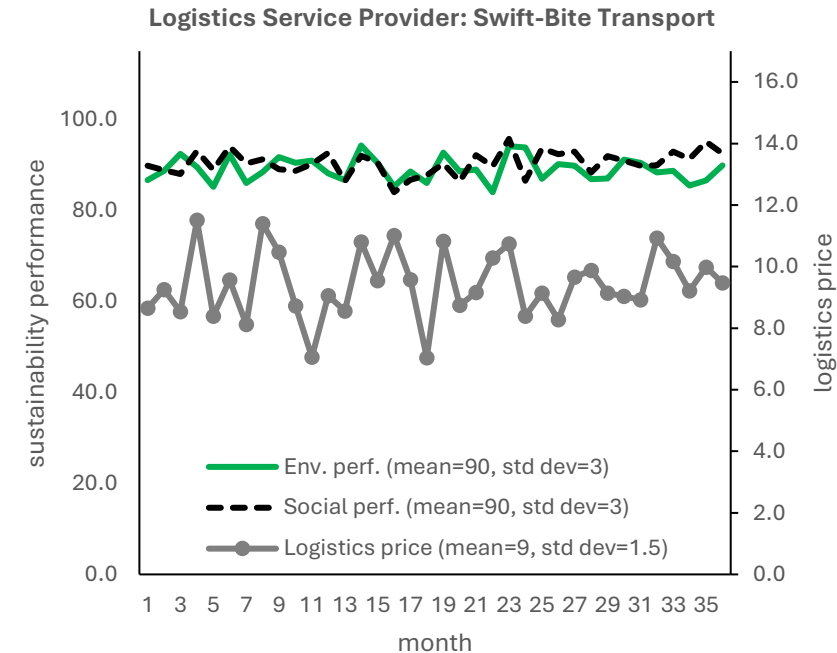


Logistics Service Provider 3: Swift-Bite Transport

- A forward-thinking enterprise



- Electric and hybrid vehicles
- Route optimisation
- Staff training
- High prices
- High sustainability performance



Logistics Service Provider 4: Daily-Haul Distribution

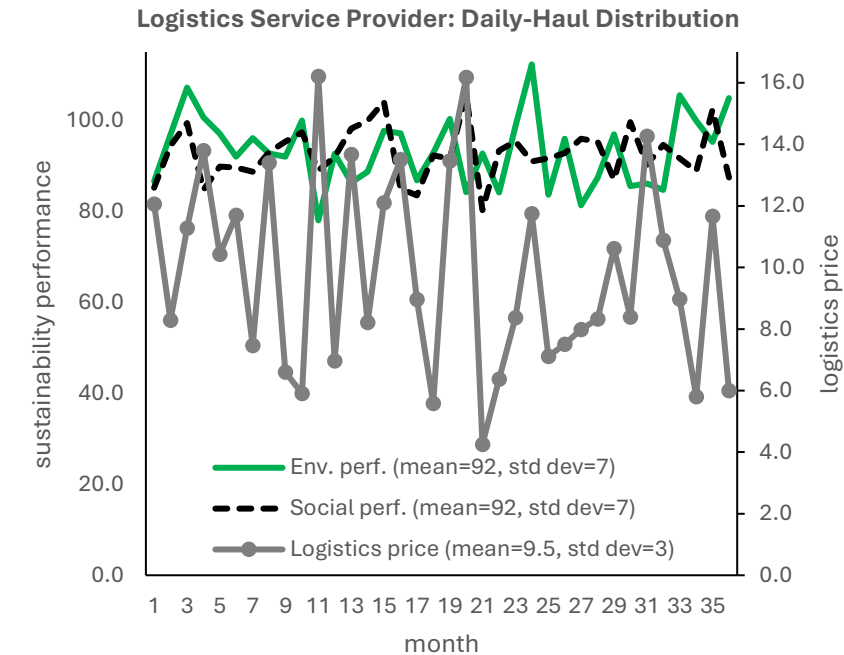


A premium logistics provider

Compared to Swift-Bite Transport

- Higher yet more unstable prices
- Higher yet less stable sustainability

- Carbon-neutral warehousing
- Advanced emissions tracking



Logistics Service Provider 5: Urban-Crate Express

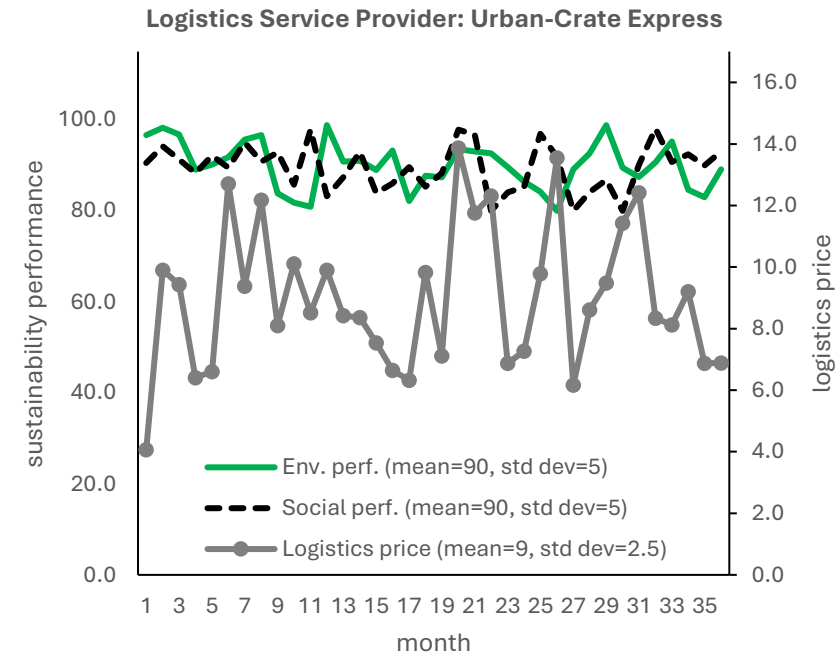


Decentralised model

- Last-mile and urban deliveries
- Electric vans and cargo bikes

Compared to Swift-Bite Transport

- Equally high yet more unstable prices
- Equally high yet more unstable sustainability



What to do next?

- Teamwork
- Analysis
- Decision-making
- Problem-solving
- Performance evaluation
- Reflection
- Planning for improvement